

EPIDEMIOLOGICAL DATA SCIENCE

Smoking Health Risk Analysis

A probabilistic framework linking patient
and demographics to evidence-based cancer
WHO, IARC, and SEER research.

30,001

PATIENT RECORDS

3

SOURCE FILES

8

SCIENTIFIC REFERENCES

4

CANCER TYPES TRACKED

// PROJECT OVERVIEW

What This Project Does

A cancer risk scoring engine that merges demographic data, smoking history, organ health, and lifestyle factors into a single probabilistic output — calibrated against peer-reviewed epidemiological literature.



Smoking Risk Quantification

Models cumulative tobacco exposure using the Pack-Years metric (Doll & Peto, 1976), scaling cancer risk by intensity and duration across multiple organ sites.



Multiplicative Risk Model

Combines independent risk factors — smoking, BMI, blood pressure, family history, alcohol — using a multiplicative epidemiological formula for biological accuracy.



Organ-Level Targeting

Assigns SEER-based baseline lifetime risks per organ and adjusts for organ condition, including a field-cancerization effect ($RR=1.6$) for chronically damaged tissue.



Evidence-Based Calibration

All Relative Risk values sourced from Gandini et al. (2008) and IARC Monograph Vol. 83, cross-validated against GLOBOCAN 2020 global incidence statistics.

Three-File Dataset Structure

Three complementary files, each capturing a distinct dimension of patient health, joined on Patient_ID.

.xlsx

Cancer Diagnosis Data

- Cancer_Diagnosis (Yes / No)
- Cancer_Type (Lung, Throat, Oral, Bladder)
- Patient_ID — join key
- Passive_Smoking status

.csv

Hospital Patient Records

- Age, Gender, BMI
- Smoking_Status, Years_of_Smoking
- Cigarettes_Per_Day
- Organ & Organ_Condition
- BP_Risk, Cholesterol_Level
- Family_History_Risk
- Alcohol_Consumption

.pdf

Scientific Reference Sheet

- WHO, IARC, NCI / SEER sources
- Gandini et al. RR values
- Doll & Peto Pack-Years formula
- GLOBOCAN cross-validation
- Multiplicative model justification

// ANALYTICAL PIPELINE

Step-by-Step Workflow

01

Data Ingestion & Merge

Load the hospital CSV and cancer diagnosis Excel, merge on Patient_ID. Validate for duplicates, missing values, and type mismatches in the lightly corrupted CSV.

02

Pack-Years Calculation

Compute cumulative tobacco exposure: $(\text{Cigarettes_Per_Day} / 20) \times \text{Years_of_Smoking}$. Stratify into Light, Moderate, Heavy, and Very Heavy bands.

03

Relative Risk Lookup

Map each patient's smoking status and organ to literature-sourced RR values (Gandini 2008, IARC Vol. 83). Apply field-cancerization multiplier for damaged organs.

04

Multiplicative Risk Score

Combine all risk factors multiplicatively using the Rothman & Greenland (1998) formula to produce a final cancer probability per patient per organ.

05

Validation Against Diagnosis

Compare model probability outputs against actual Cancer_Diagnosis labels to assess discriminative power (AUC, sensitivity, specificity).

// PROBABILISTIC MODEL

Core Risk Formula

Based on Rothman & Greenland (1998) — multiplicative epidemiological model:

```
// Final cancer probability per organ  
P = 1 - (1 - baseline_risk) ^ combined_RR  
  
// Combined Relative Risk  
combined_RR = RR_smoking × RR_organ × RR_bmi × RR_bp ×  
RR_family × RR_alcohol
```

Baseline lifetime risk & RR per organ — SEER + Gandini 2008:

Lungs

Baseline: **6.0%**
Current RR: **15.0×**
Former RR: **4.0×**

Kidney

Baseline: **1.8%**
Current RR: **1.7×**

Liver

Baseline: **1.1%**
Current RR: **1.8×**

Heart

Baseline: **0.1%**
Damaged RR: **1.6×**

Evidence Base

All risk parameters derived from peer-reviewed epidemiological literature, not assumed values.

01

WHO — Tobacco and Cancer Fact Sheet (2023)

85% of lung cancer cases attributed to tobacco — basis for Lungs high baseline risk

02

IARC Monographs Vol. 83 — Tobacco Smoke

Group 1 carcinogen. RR estimates: Liver RR=1.8, Kidney RR=1.7

03

Gandini et al. (2008) — European Journal of Cancer

Primary meta-analysis source. Lungs: Current=15.0×, Former=4.0×

04

Doll & Peto (1976) — British Medical Journal

Pack-Years formula: (Cigarettes/day ÷ 20) × Years smoked

05

NCI SEER Database — Age-Specific Cancer Incidence

Baseline lifetime risks: Lungs 6%, Kidney 1.8%, Liver 1.1%, Heart 0.1%

06

Rothman & Greenland (1998) — Modern Epidemiology

Multiplicative model: $P = 1 - (1 - \text{baseline})^{\text{combined_RR}}$

07

Slaughter et al. (1953) — Field Cancerization Theory

Damaged organ condition assigned RR=1.6

08

IARC GLOBOCAN 2020 — Global Cancer Observatory

Cross-validation of baseline incidence rates across 185 countries

Epidemiology

Cancer Risk Modeling

Smoking

Pack-Years

Multiplicative Model

SEER Data

WHO / IARC

Relative Risk

30,001 Records

Cancer Risk Probability Model — For Educational & Research Purposes Only

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